

Mathematics

A Cultural Heritage

Lecture 1

Introduction: Mayan Numbers

The End of the World

Back in 2012, many people were hysterical about the “end of the world” because the Mayan calendar had stopped on a specific day in December.

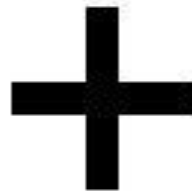


Number System

- Shells are zero
- Dots are 1
- Bars are five

0	1	2	3	4
	•	••	•••	••••
5	6	7	8	9
				
10	11	12	13	14
				
15	16	17	18	19
				

Arithmetic - Exercise



Mayan Calendars

Haab Calendar:

- 18 months, each with 20 days
- + 5 “wasted/extra” days (Uayeb/Wayeb, the last picture on the right)
- Most similar to our calendar:
 $18 \times 20 + 5 = 365$ days
- No leap years



Haab

Tzolk'in Calendar:

- Religious Calendar
- 20 months, 13 days = 260 days

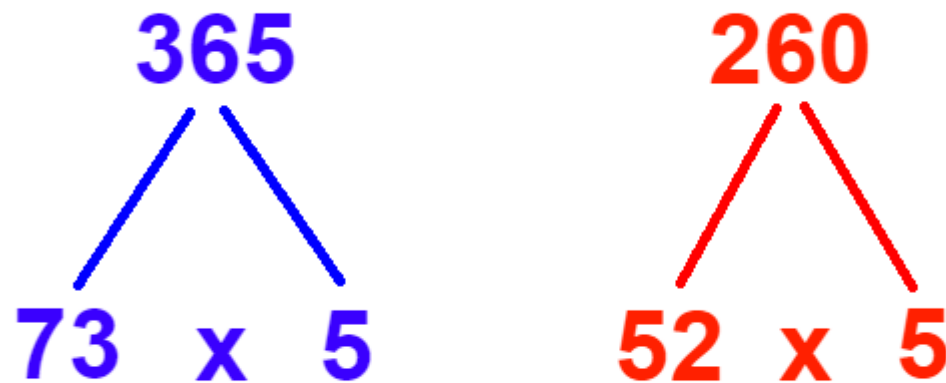
“20” is used in both calendars but in different ways.



Tzolk'in

Haab and Tzolk'in Calendars

How often would (the beginnings of the) **Haab-year (365 days)** and the **Tzolk'in (260 days)** year coincide?



Lowest Common Multiple = **73** x **52** = 18980 days, or:
73 Tzolk'in years, or 52 Haab years

These were called **calendar rounds**.